

Course Title: System Analysis and Design | Quiz 3 | Course Code: CSE321 | Time: 45 Minutes | Total Marks: 15 | Section: RE_A(3C)

Question 1 — Project Management & Gantt Chart [5 Marks]

Task ID	Task Name	Duration (Weeks)	Predecessor
A	Requirements Analysis	2	—
B	Database Design	3	A
C	User Interface Design	2	A
D	Backend Development	4	B
E	Frontend Development	3	C
F	Search & Recommendation Module	3	D
G	Integration & Unit Testing	2	D, E
H	System Testing	2	F, G
I	User Training & Documentation	1	H
J	Deployment	1	I

(a) Draw a Gantt Chart for the DLMS project showing all tasks, their durations, dependencies, and timeline. Use proper bars, labels, and timeline markings (weeks 1–20). **[3 Marks]**

(b) Identify the Critical Path of this project. List all tasks in the critical path sequence and calculate the minimum project completion time. Show all calculation steps. **[2 Marks]**

Question 2 — Information Security & Cybercrime Analysis [5 Marks]

Scenario:

Rashed, a computer science student at DIU, received an email claiming to be from the university's IT department stating that his student portal password would expire within 24 hours. The email contained an urgent message with a link to 'update credentials immediately.' The link directed to a website that looked identical to the official DIU portal. Rashed entered his student ID, password, and personal information including his phone number and date of birth.

Three days later, Rashed discovered that:

- Someone had logged into his student portal and changed his course registration.
- His Facebook account (same password) was posting spam messages.
- He received calls from unknown numbers offering fake job opportunities using his personal data.
- University server logs showed multiple failed login attempts from different IP addresses using his credentials.

Meanwhile, investigation revealed that several lab computers were infected with malicious software that was:

- Recording all keystrokes typed by students.

- Capturing screenshots every 30 seconds.
- Sending collected data to an external server in another country.
- Using infected computers to send thousands of fake requests to the admission portal, causing it to slow down significantly.

Questions

(a) Identify and explain FOUR different types of cybercrimes or security threats present in this scenario. For each threat, provide: The specific name of the cybercrime/threat, A brief explanation of how it works, Evidence from the scenario that supports your identification. **[2.5 Marks]**

(b) The malicious software found on lab computers exhibits characteristics of multiple malware types. Identify TWO specific types of malware based on their behaviors described and explain their primary functions. **[2.5 Mark]**

Question 3 — System Testing Methodology [5 Marks]

Issue	Description
Issue 1	500 students simultaneously searching for books during exam week, causing the system to take 45 seconds instead of 3 seconds for search results.
Issue 2	A student can enter SQL commands in the Book Search field (' OR '1'='1), exposing restricted administrative records.
Issue 3	When the database server is restarted unexpectedly during a book borrowing transaction, the system shows incorrect book availability status.
Issue 4	The fine calculation module produces incorrect results: book returned 5 days late at 10 Tk/day calculates 47 Tk instead of the correct 50 Tk.
Issue 5	The system integrates successfully with the university's student information system but fails to connect with the third-party payment gateway.
Issue 6	After adding the new 'E-book Download' feature, the previously working 'Book Reservation' feature stops functioning correctly.
Issue 7	The user manual states students can reserve up to 5 books simultaneously, but testing shows the system allows only 3 reservations.

Questions

(a) For EACH of the seven issues listed above, identify the most appropriate type of system testing that should be performed to detect and resolve the problem. Justify your answer. **[2.5 Marks]**

(b) The QA team needs to test the 'Book Borrowing' functionality. They have access to the system's source code and database structure but want to focus on testing user-facing functionality. Which testing approach is most suitable? Justify your choice? Design a test case for: "A student attempts to borrow a book that is already borrowed by another student." Your test case must include: Test Case ID, Pre-conditions, Test Steps (minimum 4), Test Data, Expected Result, Actual Result, and Status.? **[2.5 Marks]**